ROBERTO MARTÍN GUTIÉRREZ

AI ENGINEER MEDICAL AI · COMPUTER VISION

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PROFILE

Biomedical Engineer with an MSc (University of Valladolid). Developing **healthcare AI**, computer vision, LLMs and clinical software, with experience in robotics, deep learning and cardiovascular trials.

EXPERIENCE

AI & Computational Cardiology Engineer
ICICOR · HCUV | MAR 2026 – PRESENT

Developing and evaluating AI for **echocardiography, signals and clinical text**; integrating models, APIs and research software. Supporting clinical trials.

Robotics Technical Lead
ASTIBOT | FEB – MAR 2026

Coordinated **software and electronics** teams, AI–hardware integration, UVa internship supervision and European R&D proposals.

AI & Robotics R&D Programmer
ASTIBOT | OCT 2025 – FEB 2026

Computer vision using **Python, ROS 2 and Docker**; RGB-D/LiDAR perception and ROVER X / TrimBot systems integration.

Biomedical AI Research · MSc Thesis
UVa / ISCIII COLLABORATION | JAN – JUN 2025

Virtual staining using **U-Net / ResNet**; Keras 3–bioMONAI/MONAI interoperability on PyTorch. **Thesis: 9.4/10**.

Biomedical Research · BSc Thesis / Intern
IBGM · CSIC / UVa | JAN – JUL 2024

CRISPR/Cas9 mitochondrial disease cell models and computational support for genetic model analysis. **Thesis: 9.3/10**.

SELECTED PROJECTS

- **ECOCARDIA** · Echo AI: YOLO, 3D CNN, TimeSformer, Qwen-VL. ISCIII DTS25/00100.
- **EchoExtract-ETT** · LLM / NLP pipeline for structured extraction from clinical reports.
- **AnonimICICOR** · Offline clinical anonymization tool; backend, frontend and batch flows.
- **EMBOENDO-DL** · Supporting the AI team on vegetation imaging and embolic risk. PI23/01072.
- **CaixaImpulse CI24-10743** · Research team member: AI for embolism prediction in endocarditis.
- **CONVALV** · Heart sound classifier: CNN, Mel spectrograms and REST API.

CLINICAL TRIALS

- **START YOUNG** · International TAVI vs. surgery trial. NCT06861361.
- **VitaFlow LIBERTY Europe** · Multicenter TAVI valve study. NCT06535659.

SCHOLARSHIPS & AWARDS: **Santander–UVa Excellence Scholarship** (2024) · High School Academic Honors (2020).

LANGUAGES: Spanish native · English limited working · German elementary · Driving licence B

TECH STACK

AI / Deep Learning
Python · PyTorch · TensorFlow · Keras · Transformers

LLMs / RAG
RAG · LangChain · FAISS · embeddings · Qwen-VL · local LLMs

Computer Vision
YOLO · OpenCV · U-Net · ResNet · 3D CNN · TimeSformer · MONAI

Backend / APIs
FastAPI · Flask · REST · OpenAPI · Uvicorn · SQLite

Frontend / DevOps
React · Vite · Next.js · Docker · Git/GitHub · Linux · Nginx

Robotics / data
ROS 2 · SLAM · LiDAR · RGB-D · MoveIt · Nav2 · Gazebo · MATLAB · C/C++

AWS / Amazon Bedrock
Introductory AWS and Bedrock training (2026).

EDUCATION

MSc Biomedical Engineering
UVa · 2024–2026 | Thesis 9.4/10

BSc Biomedical Engineering
UVa · 2020–2024 | Thesis 9.3/10

Science & Technology High School
IES Julián Marías · 2018–2020 | 9.58/10

COURSES & CERTIFICATES

AWS: Amazon Bedrock Getting Started (2026)

IBM: RAG for Enhanced AI Outputs (2026)

Hugging Face: AI Agents Fundamentals (2026)

Kaggle: Computer Vision (2026)

Global Health Network: ICH Good Clinical Practice E6(R3) (2026)

ACCIS: Cardiology meeting · 14 hours (2026)

UVa: External Internship Supervisor (2026)

Anthropic: Claude Code in Action · Subagents · Claude Code 101 (2026)